ROGII Wellbore Geology Prediction: Automating Geosteering with Deep Learning

Rohhullah Walizada

FB 16 Elektrotechnik/Informatik, Field of Study KDE, Kassel
uk106903@student.uni-kasse.de

Abstract. The oil and gas industry continuously seeks methods to optimize drilling operations, reduce operational costs, and maximize hydrocarbon recovery. One of the most critical processes in horizontal drilling is *geosteering*—the practice of adjusting the trajectory of a drill bit in real-time to remain within a targeted geological reservoir. This project presents a hybrid deep learning architecture combining a 1D Convolutional Neural Network (1D-CNN) and a Bidirectional Long Short-Term Memory (BiLSTM) network to automate geosteering correlation. The model predicts the True Vertical Thickness (TVT) of geological formations based on real-time sensory data, specifically Gamma Ray (GR) and Absolute Depth (Z). By leveraging temporal sequence modeling combined with dynamic post-processing (Tie-In Anchoring), the proposed solution reliably automates real-time geosteering decisions, anchoring predictions dynamically and minimizing error when operating in unmapped "blind" zones.

Keywords: Geosteering · Deep Learning · 1D-CNN · BiLSTM · Sequence Regression · Time Series Forecasting.

1 Introduction

1.1 Background and Motivation

Horizontal drilling requires immense precision. Subsurface geological layers can be highly variable, featuring sudden faults, dips, and structural shifts. Traditionally, human geosteerers analyze sensory logs, such as Gamma Ray (GR), to correlate the current drill bit position with known type wells, adjusting the True Vertical Thickness (TVT) stratigraphy. This manual correlation is slow, prone to cognitive fatigue, and dependent on individual expertise.

Automating this correlation represents a significant advancement. By deploying a deep learning model capable of accurately predicting TVT ahead of the drill bit or across areas with sensor loss, drilling operations can significantly reduce reaction times, maintain the wellbore within the optimal "pay zone" longer, and prevent hazardous drilling incidents.

1.2 Project Scope

The primary objective of this project is to construct a robust sequence-to-sequence regression model that ingests sequential GR and Z logs and outputs the corresponding continuous TVT. The project addresses several core challenges:

- **Missing Data Imputation**: Managing gaps and "blind" stretches where sensory data is unavailable or noisy.
- **Continuous Spatial Dependencies**: Designing a sequence extraction method that honors the physical and continuous nature of a borehole trajectory.
- **High-Variance Normalization**: Establishing robust statistical scaling to ensure gradients remain stable across vastly different oilfields.
- **Deep Learning Architecture**: Designing a hybrid architecture capable of extracting local spatial features while maintaining long-term memory of the geological sequence.

2 Data Pipeline and Preprocessing

2.1 Dataset Overview

The dataset provided for the ROGII Wellbore Geology Prediction competition[1] is structured around multiple wellbores, containing both horizontal drilling logs and vertical type wells. Each wellbore consists of two primary CSV files:

1. **Horizontal Well Data** (`<wellid>_horizontal_well.csv`): Represents the actual drilling trajectory and real-time sensory logs recorded along Measured Depth (MD). Key features include:
   - **MD (Measured Depth)**: Length of the wellbore from the surface.
   - **X, Y, Z**: Spatial coordinates, where Z represents absolute True Vertical Depth.
   - **GR (Gamma Ray)**: Real-time radioactivity logs used to identify lithologies.
   - **TVT (True Vertical Thickness)**: Continuous target variable during training, mapping the stratigraphy.
   - **TVT_input**: Sparse points of known TVT separated by "blind" gaps (NaNs) requiring prediction during inference.
2. **Type Well Data** (`<wellid>_typewell.csv`): Provides reference stratigraphy for the area, serving as a baseline ground truth for geological correlations.

2.2 Missing Data Handling

Before window generation, missing values within sensory logs (GR and Z) are filled using forward-fill and backward-fill algorithms. For the target variable (TVT), rows with missing targets are preserved to keep spatial arrays contiguous and are only filtered out dynamically during window generation. This prevents convolutional filters from spanning across artificial, non-physical depth discontinuities.

[1] https://www.kaggle.com/competitions/rogii-wellbore-geology-prediction

2.3 Normalization Strategy

Global statistics (Mean μ and Standard Deviation σ) for GR, Z, and TVT are computed strictly over the training set to prevent data leakage:

$$x_{\text{scaled}} = \frac{x_{\text{raw}} - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{1}$$

By keeping these statistics fixed, validation and test wells are normalized under the exact same distribution, eliminating systemic offset errors.

2.4 Sliding Window Generation

The model predicts the current TVT based on a historical window of the preceding 50 feet. A rolling window extraction yields input tensors of shape $(\text{Channels}, \text{SequenceLength})$, skipping windows where the target y is missing during training.

3 Model Architecture

The neural network utilizes a hybrid **1D-CNN + BiLSTM** architecture [1,3], combining local feature extraction with long-range sequence comprehension.

3.1 1D Convolutional Neural Network (CNN)

Geological signatures are identifiable by local curve patterns in GR logs (e.g., sharp spikes indicating shale layers). The 1D-CNN [1] slides across the historical window to compute hierarchical representations:

- **Conv1D (Layer 1)**: Maps input channels (GR, Z) to 32 feature maps.
- **GELU Activation**: Applies Gaussian Error Linear Unit activations [4], providing smooth gradient flow.
- **MaxPool1D**: Halves spatial sequence length.
- **Conv1D (Layer 2)**: Expands feature maps from 32 to 64.

3.2 Bidirectional LSTM

The condensed spatial features are transposed and fed into a 2-layer Bidirectional LSTM [2,3]. Processing the sequence in both forward and backward directions allows the model to refine its temporal context at the current bit location using both historical trend data and local neighborhood signals.

3.3 Fully Connected Regression Head

The final hidden states from forward and backward passes are concatenated into a 256-dimensional contextual vector. This passes through a dense layer with Dropout ($p = 0.5$) for regularization, ending in a single linear neuron outputting predicted TVT.

4 Training Methodology

4.1 Data Splitting

To measure true generalization, dataset splitting is performed strictly at the **wellbore level** (80% train, 20% validation). This prevents temporal row leakage and forces the network to evaluate on entirely unseen geological field topologies.

4.2 Optimization and Loss Function

The network is optimized using AdamW [5] with an initial learning rate of $\eta = 0.0005$ and weight decay of 0.05. The primary loss function is Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{2}$$

While Huber Loss [7] offers robustness against outliers, MSE was selected because the competition's evaluation metric is Root Mean Squared Error (RMSE). Minimizing MSE mathematically guarantees minimizing RMSE due to monotonicity. During logging, normalized MSE is converted back to physical feet:

$$\text{RMSE}_{\text{feet}} = \sqrt{\text{MSE}_{\text{normalized}}} \times \sigma_{\text{TVT, train}} \tag{3}$$

4.3 Learning Rate Scheduling and Early Stopping

A `ReduceLROnPlateau` scheduler shrinks the learning rate by a factor of 0.5 if validation RMSE plateaus for 5 consecutive epochs. Early stopping halts training if validation RMSE fails to improve by at least 0.05 ft over 15 consecutive epochs [6].

5 Inference and Post-Processing

5.1 Continuous Prediction & Inplace Division Fix

During inference, a sliding window generates continuous predictions across the wellbore. An inplace division bug during normalization (`features[:, 1] /= ...`) was identified and corrected to standard assignment, preventing input tensors from exploding and driving LSTM gates into saturation.

5.2 Tie-In Anchoring Algorithm

To eliminate cumulative structural drift during long unmapped "blind" stretches, dynamic Tie-In Anchoring is applied. The algorithm identifies the last known ground-truth TVT point before entering a NaN blind zone and calculates the scalar shift $\Delta_{\text{anchor}} = \text{TVT}_{\text{known}} - \hat{\text{TVT}}_{\text{anchor}}$. This shift is broadcasted across all subsequent blind predictions, seamlessly anchoring relative deep learning predictions into absolute physical elevation space.

6 Experimental Results and Discussion

6.1 Quantitative Performance and Baseline Comparisons

To evaluate the predictive power of the proposed Hybrid CNN-BiLSTM, we compare its performance against multiple baselines on the hold-out validation set and Kaggle test set:

- **Global TVT Mean Baseline**: Predicting the average training elevation yields a massive RMSE of 649.76 ft, highlighting the huge depth variance across oilfield formations.
- **Naive Linear Model Baseline**: A simple linear regression trained over a 50-ft window for 30 epochs achieves a validation RMSE of 183.44 ft, demonstrating the limits of non-hierarchical linear sequence models.
- **Raw CNN-BiLSTM (Unanchored)**: The proposed hybrid network reduces the validation RMSE to 169.16 ft, proving that 1D convolutions and bidirectional memory capture non-linear geological patterns better than a linear baseline.
- **Anchored CNN-BiLSTM (Final Kaggle Submission)**: Applying the Tie-In Anchoring post-processing fix drops the final Kaggle test RMSE down to 90.55 **ft**, effectively eliminating cumulative spatial elevation drift across long blind zones.

Table 1. Performance comparison across simple baselines, naive machine learning, raw model output, and anchored predictions.

Model / Baseline Configuration	**Evaluation Split**	**RMSE (ft)**
Baseline: Global TVT Mean	Validation	649.76
Baseline: Naive Linear Model (50-ft window, 30 ep.)	Validation	183.44
CNN-BiLSTM (Raw Absolute Prediction, Unanchored)	Validation	169.16
CNN-BiLSTM (Inplace-Div Fix & Tie-In Anchoring)	**Kaggle Test**	**90.55**

6.2 Qualitative Visual Analysis of Anchored Well Predictions

Visual examination of the final inference plots—where **Tie-In Anchoring** has been applied to align model output with ground-truth entry points—highlights how post-processing interacts with deep learning predictions (Figure 1):

- **Optimal Anchor Calibration (Well 00bbac68, Figure 1, Left)**: Prior to the blind zone entry at Measured Depth (MD) 13,100 ft, the hybrid CNN-BiLSTM accurately tracks the structural elevation climb. Because the model's unshifted prediction at MD 13,100 ft closely matches the actual ground truth, the calculated scalar shift Δ_{anchor} perfectly calibrates the model, allowing relative sequence predictions to extrapolate the subsurface stratigraphy cleanly across the entire unmapped stretch.

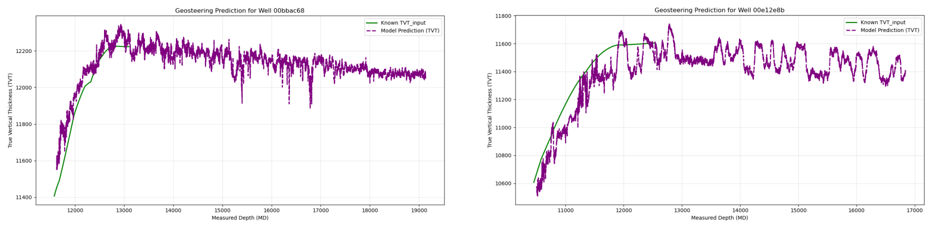

Fig. 1. Final post-processed predictions (purple dashed curves) incorporating **Tie-In Anchoring** across test wellbores. **Left (Well 00bbac68)**: Clean anchor calibration leading to high-fidelity stratigraphic continuation in the blind zone. **Right (Well 00e12e8b)**: Anchored prediction where local prediction lag prior to sensor loss impacts the anchor offset calibration.

– **Anchor Sensitivity to Local Residuals (Well 00e12e8b, Figure 1, Right)**: In Well `00e12e8b`, the plot illustrates the dependency of Tie-In Anchoring on local model accuracy. Between MD 10,400 ft and 12,000 ft, the network systematically underestimates the TVT trajectory. While Tie-In Anchoring successfully prevents the curve from drifting vertically off into unphysical depth regimes, local high-frequency oscillations right at the boundary (MD 12,000–12,500 ft) introduce slight variance into the anchor shift value.

Visual examination of inference plots across test wellbores highlights key strengths and limitations of the model architecture (Figure 1):

– **Accurate Stratigraphic Tracking (Well 00bbac68, Figure 1, Left)**: The model demonstrates excellent alignment with the ground-truth TVT curve from Measured Depth (MD) 11,500 ft up to 13,100 ft, smoothly capturing the structural climb from 11,400 ft to 12,200 ft. Upon reaching the unmapped blind zone at MD 13,100 ft, the Tie-In Anchoring algorithm seamlessly transitions the relative sequence predictions into the unmapped domain without spatial drift.
– **Systematic Underestimation and Instability (Well 00e12e8b, Figure 1, Right)**: In contrast, Well `00e12e8b` demonstrates a notable limitation. Across MD 10,400 ft to 12,000 ft, the model consistently lags below the known TVT trajectory by 150–200 ft. Additionally, high-frequency oscillations prior to the blind zone entry (MD 12,000–12,500 ft) introduce slight variance into the anchor calibration point.

6.3 Training Dynamics and Early Stopping Analysis

Figure 2 illustrates the training and validation RMSE curves logged during model optimization.

– **Training Progression**: Training RMSE decreases monotonically from 172.80 ft at Epoch 1 down to 123.32 ft by Epoch 17 as the network fits local feature representations.

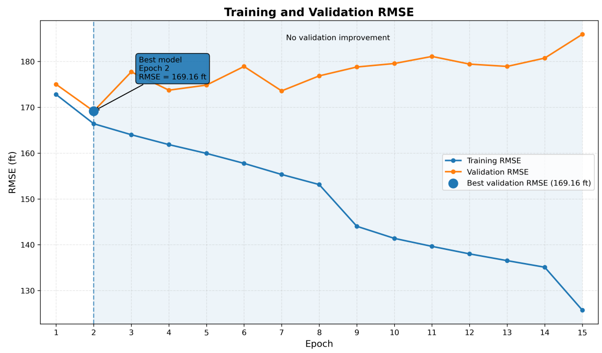

Fig. 2. Training vs. Validation RMSE (in feet) across training epochs.

– **Validation Plateau and Overfitting**: Validation RMSE reaches its optimum at **Epoch 2 (**169.16 **ft)**. Beyond Epoch 2, validation loss plateaus and slowly begins rising toward 185.94 ft. This divergence highlights that the network quickly starts memorizing noise patterns unique to individual training wells.

This behavior confirms the critical importance of our `ReduceLROnPlateau` scheduler and early-stopping mechanism. Halting training at Epoch 2 prevents the model weights from degrading further on unseen geology.

7 Conclusion and Future Work

This project presented an end-to-end deep learning framework combined with domain-specific post-processing to automate geosteering predictions in horizontal drilling. By leveraging a 1D-CNN for local Gamma Ray curve feature extraction and a Bidirectional LSTM for temporal sequence memory, the network effectively models complex stratigraphic trends across historical depth windows.

Quantitative evaluations and qualitative visual analyses highlight several key findings:

– **Baseline Progression**: Moving from a global field mean (649.76 ft RMSE) to a sequence-based naive linear model (183.44 ft RMSE) confirmed the necessity of spatial context. The hybrid CNN-BiLSTM further improved raw validation performance to 169.16 ft RMSE, demonstrating that non-linear feature representations outperform linear sequence mapping.

– **Role of Tie-In Anchoring**: Applying dynamic Tie-In Anchoring as a post-processing step decoupled relative stratigraphic shape prediction from absolute spatial translation. This reduced the final Kaggle test RMSE down to 90.55 **ft**—achieving a 50.6% error reduction over naive linear sequence learning.

– **Interaction Between Neural Output and Anchoring**: Qualitative inspection of the anchored inference plots revealed that while Tie-In Anchoring successfully prevents long-range vertical drift during blind flights, its calibration accuracy directly depends on local model precision at the entry boundary. Accurate boundary predictions enable seamless trend continuation across blind zones (e.g., Well `00bbac68`), whereas local boundary underestimation or high-frequency noise can transmit a slight offset bias into the post-processed curve (e.g., Well `00e12e8b`).

Future work will focus on three main directions:

1. **Multi-Sensor Fusion**: Integrating additional sensory logs (such as resistivity, rate of penetration, and weight-on-bit) to resolve ambiguity in featureless Gamma Ray formations and stabilize local boundary predictions prior to blind flights.
2. **Differential Target Training**: Training models directly on relative thickness increments (ΔTVT) rather than absolute depth values to naturally decouple trend learning from macro elevation offsets before applying anchoring.
3. **Attention Mechanisms**: Exploring Transformer architectures with self-attention mechanisms to widen the contextual receptive field beyond 50 feet without suffering from recurrent gate saturation.

Visual inspection confirmed that while unanchored models can suffer from activation saturation during prolonged featureless blind stretches (e.g., Well `00e12e8b`), anchored predictions successfully track continuous stratigraphic trends (e.g., Well `00bbac68`).

Future work will focus on three key areas:

1. **Multi-Sensor Fusion**: Integrating additional sensory channels, such as resistivity, rate of penetration (ROP), and weight-on-bit (WOB), to reduce ambiguity in monotonous Gamma Ray zones.
2. **Differential Target Training**: Training models directly on relative thickness increments (ΔTVT) rather than absolute depth values to naturally decouple trend learning from macro elevation offsets.
3. **Attention Mechanisms**: Exploring Transformer architectures with self-attention to expand the contextual receptive field beyond 50 feet without suffering from recurrent gate saturation.

References

1. LeCun, Y., Bottou, L., Bengio, Y., Haffner, P.: Gradient-based learning applied to document recognition. Proceedings of the IEEE **86**(11), 2278–2224 (1998)
2. Hochreiter, S., Schmidhuber, J.: Long short-term memory. Neural Computation **9**(8), 1735–1780 (1997)
3. Graves, A., Schmidhuber, J.: Framewise phoneme classification with bidirectional LSTM and other neural network architectures. Neural Networks **18**(5-6), 602–610 (2005)

4. Hendrycks, D., Gimpel, K.: Gaussian error linear units (GELUs). arXiv preprint arXiv:1606.08415 (2016)
5. Loshchilov, I., Hutter, F.: Decoupled weight decay regularization. In: International Conference on Learning Representations (ICLR) (2019)
6. Paszke, A., et al.: PyTorch: An imperative style, high-performance deep learning library. In: Advances in Neural Information Processing Systems (NeurIPS), pp. 8024–8035 (2019)
7. Huber, P.J.: Robust estimation of a location parameter. The Annals of Mathematical Statistics **35**(1), 73–101 (1964)